

LAB 1 ASSIGNMENT

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1 Introduction

The term Heuristics comes from the expression "Eureka!"¹ and that is what Archimedes said when he developed his Fluids Theorem. By their name we can imagine that Informed Strategies are going to be better than Uninformed Strategies since the first ones are going to be informed. Informed Strategies are more efficient since they are more informed about the problem, they can weight different decisions and, thus, take the optimal decision.

2 Statistical Analysis

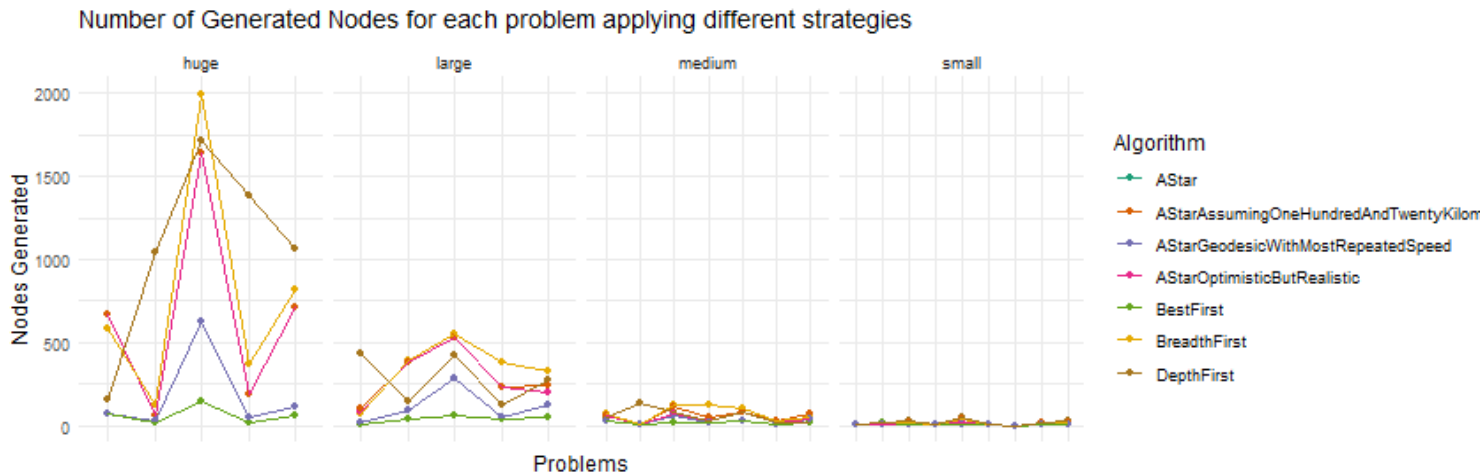


Figure 1: Graphic showing the nodes generated of each strategy within each dimension of the problems.

In the above graphic we can see how all the strategies generate the same number of nodes in small problems. Nevertheless, as the dimension of the problems grow, each strategy generate a different number of nodes. Note that the BestFirst strategy is the best one since it generates almost none nodes compared to BreadthFirst or DepthFirst.

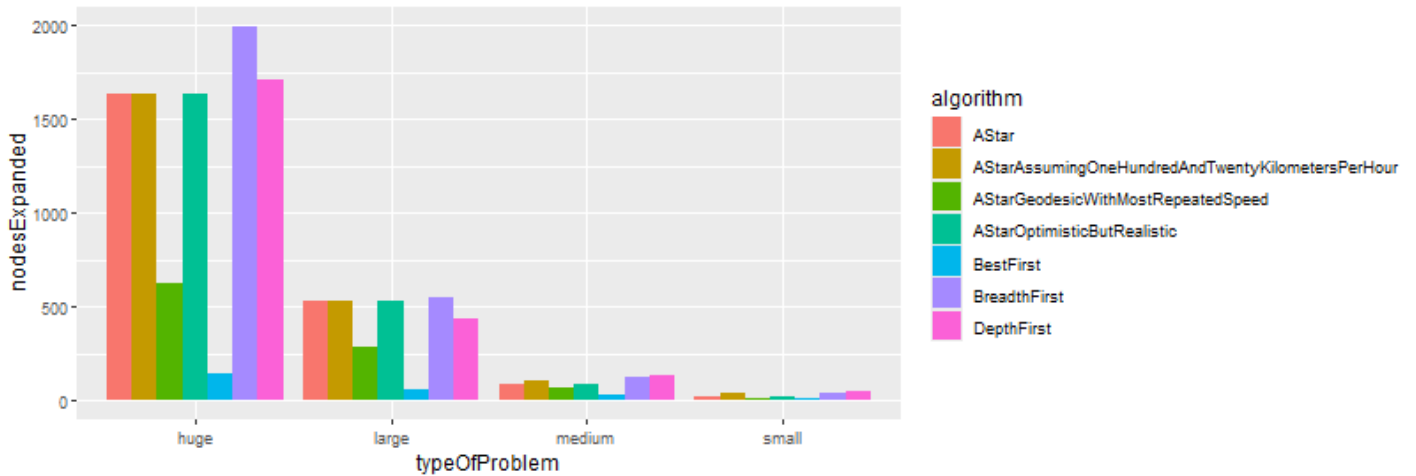


Figure 2: Graphic showing the nodes expanded of each strategy within each dimension of the problems.

In this other graphic we can appreciate the number of nodes that are expanded by each search strategy within each dimension of the problem. We can see how in small problems we can't really see the difference but if the problem grows the search strategies that are most dominant (by not expanding nodes almost) are the **BestFirst** and **AStarGeodesicWithMostRepeatedSpeed**. It's interesting to mention that **AStarGeodesicWithMostRepeatedSpeed** strategy beats most of the times **AStar** and anyone else in terms of generated nodes, expanded nodes and depth of solution having a little bit more of solution cost than other strategies (insignificant).

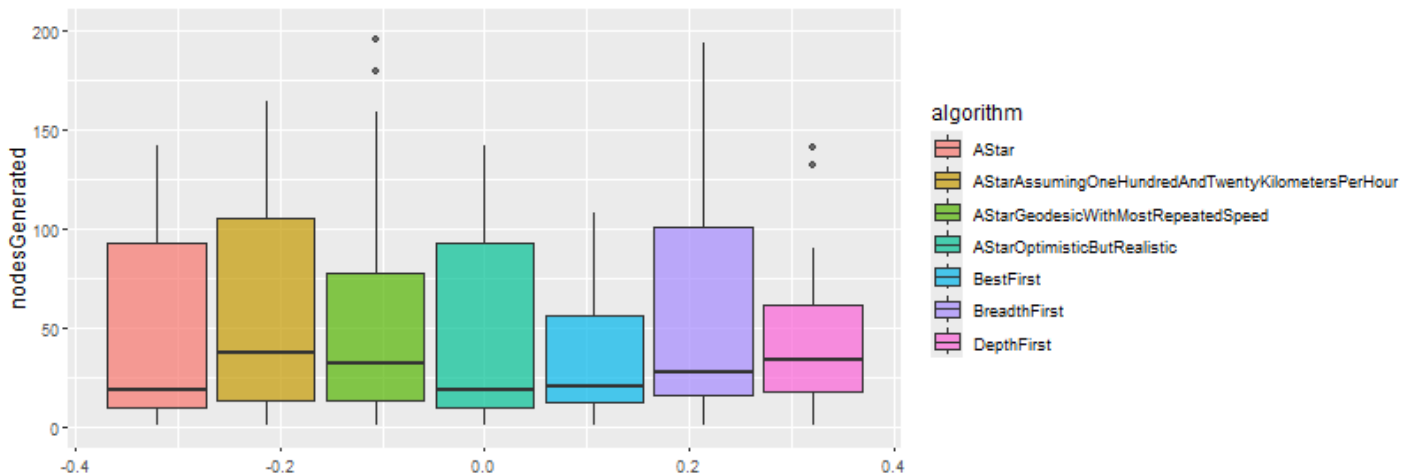


Figure 3: Boxplot on nodes generated by each search strategy.

3 Algorithmic complexity

identifying the key elements we have been working with in class:

- **Solution Data Type:** The solution is a mapping of goods to trucks, where each truck's weight limit is not exceeded.

¹Artificial Intelligence: A Modern Approach. Russell, Stuart J.

- **Exhaustivity:** The algorithm explores all possible combinations of assigning goods to trucks until a feasible solution is found or all combinations are exhausted.
- **Dead Node Condition (Backtracking):** If placing a good in a truck would exceed its capacity, the algorithm backtracks to the previous state and explores other possibilities.
- **Live Node Condition:** If the current assignment of goods to trucks does not exceed any truck's capacity and there are goods remaining, the algorithm continues exploring further assignments.
- **Solution Node Condition:** If all goods are successfully assigned to trucks without exceeding any capacity, a solution is found.

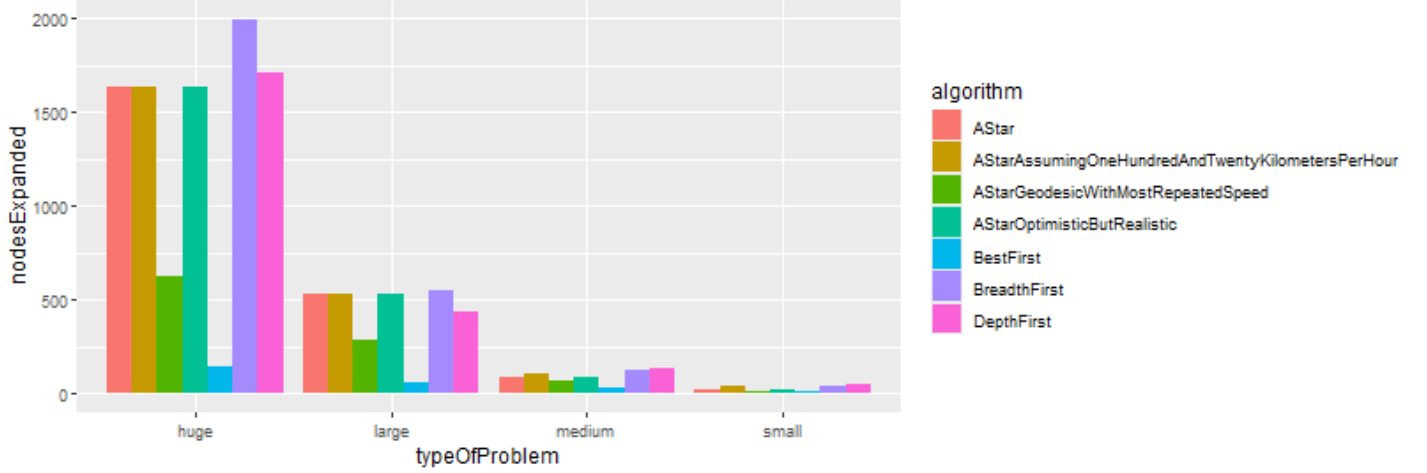


Figure 4: Cost of each strategy used for problems with different dimensions.

3.0.1 Optimization

The following optimizations have been applied to improve the efficiency of the backtracking algorithm:

1. **Feasibility Check:** Before starting the backtracking process, a feasibility check is performed to determine if it's even possible to distribute all goods among the given number of lorries without exceeding the capacity of each lorry or the budget constraint.
2. **Sorting:** The weights of goods are sorted in non-increasing order using the `quickSort` function. Sorting the weights allows the algorithm to try heavier goods first, potentially reducing the search space.
3. **Pruning:** During the backtracking process, if a particular combination of goods and lorries exceeds the load capacity of any lorry, that branch of the search tree is pruned. Similarly, if the cost exceeds the budget constraint, the branch is also pruned. This avoids exploring solutions that are guaranteed to be invalid.
4. **Array to Track Goods Assignment:** An array `selected[NUM_LORRIES][n]` is used to keep track of which goods are assigned to which lorries. This allows the algorithm to print out the weights assigned to each lorry once a valid solution is found.
5. **Avoiding Redundant Permutations:** When checking if a lorry has a load capacity of 0, it breaks the loop to avoid redundant permutations. This helps in reducing unnecessary computations.

3.1 Pseudocode

Note: On **Algorithm 3** *FeasibilityCheck Algorithm* we have got an implicit conversion from integer, n , to boolean. We must mention that 0 is false, true otherwise.

Algorithm 1 Main Algorithm

```

1: function MAIN
2:   Create DataBase
3:   for  $i \leftarrow small$  to  $huge$  do
4:     for  $i \leftarrow 0$  to  $numberFiles - 1$  do
5:       read  $problemNameFile$ 
6:       declare  $problem$ 
7:       declare  $searchStrategy$ 
8:       SEARCH
9:       write results on the DataBase
10:    end for
11:  end for
12: end function

```

Algorithm 2 Search Algorithm

```

1: function SEARCH( $StrategyToUse$ )
2:   if  $index = n$  then
3:     return true
4:   end if
5:    $i \leftarrow index \% 3$ 
6:   for  $j \leftarrow 0$  to  $NUM\_LORRIES - 1$ ,  $j \leftarrow j + 1$  do
7:     if  $lorries[i] + weights[index] \leq max\_capacity$  then
8:        $lorries[i] \leftarrow lorryes[i] + weights[index]$ 
9:       if CANPLACEGOODSNAKE( $weights[]$ ,  $lorries[]$ ,  $index + 1$ ,  $n$ ,  $max\_capacity$ ,  $selected[][]$ ) then
10:         $selected[i][index] \leftarrow true$ 
11:        return true
12:      end if
13:       $lorries[i] \leftarrow lorryes[i] - weights[index]$ 
14:    end if
15:    if  $lorries[i] = 0$  then
16:      break
17:    end if
18:     $i \leftarrow INCREMENT\_SNAKE(i, index)$ 
19:  end for
20:  return false
21: end function

```

Algorithm 3 FeasibilityCheck Algorithm

```

1: function FEASIBILITYCHECK( $weights[], n$ )
2:    $sum \leftarrow 0$ 
3:   for  $, n$  and  $sum \leq NUM\_LORRIES \times max\_capacity$  and  $weights[n - 1] \leq max\_capacity$  and  $sum \times cost\_per\_kg \leq budget$ , do
4:      $n \leftarrow n - 1$ 
5:      $sum \leftarrow sum + weights[n]$ 
6:   end for
7:   return not  $n$  and  $sum \leq NUM\_LORRIES \times max\_capacity$  and  $sum \times cost\_per\_kg \leq budget$ 
8: end function

```

Algorithm 4 CanPlaceGoodsBalancedSum Algorithm

```

1: function CANPLACEGOODSBALANCEDSUM(weights[], lorries[], index, n, max_capacity, selected[])
2:   if index = n then
3:     return true
4:   end if
5:   Define start and increment as integers
6:   if lorries[0] ≤ lorries[1] & lorries[0] ≤ lorries[2] then
7:     start ← 0
8:     if lorries[1] ≤ lorries[2] then
9:       increment ← 1
10:    else
11:      increment ← -1
12:    end if
13:  else
14:    if lorries[1] ≤ lorries[0] & lorries[1] ≤ lorries[2] then
15:      start ← 1
16:      if lorries[0] ≤ lorries[2] then
17:        increment ← 1
18:      else
19:        increment ← -1
20:      end if
21:    else
22:      start ← 2
23:      if lorries[0] ≤ lorries[1] then
24:        increment ← 1
25:      else
26:        increment ← -1
27:      end if
28:    end if
29:  end if
30:  i ← start
31:  for j ← 0 to NUM_LORRIES - 1, j ← j + 1 do
32:    if lorries[i] + weights[index] ≤ max_capacity then
33:      lorries[i] ← lorries[i] + weights[index]
34:      if CANPLACEGOODSBALANCEDSUM(weights[], lorries[], index + 1, n, max_capacity, selected[])
35:        selected[i][index] ← true
36:        return true
37:      end if
38:      lorries[i] ← lorries[i] - weights[index]
39:    end if
40:    if lorries[i] = 0 then
41:      break
42:    end if
43:    i ← INCREMENT_SUM(i, increment)
44:  end for
45:  return false
46: end function

```

3.2 Complexity analysis

- Analysis of the **computational complexity** of the algorithm in terms of the number of operations executed by the algorithm as a function of the input size n .

Answer: The backtracking algorithm's complexity is analyzed based on the number of operations it executes, which is a function of the input size n , the maximum load capacity of the lorries $MAX_CAPACITY$, and the maximum number of lorries $NUM_LORRIES$. The algorithm consists of several key components, each contributing to the overall complexity as follows:

1. Sorting the Weights: The algorithm begins by sorting the weights in non-increasing order. This sorting operation typically takes $O(n \log n)$ time using efficient sorting algorithms like quicksort or mergesort.
2. Feasibility Check: Before initiating the backtracking process, a feasibility check is performed to determine if it's possible to distribute all goods among the given number of lorries without exceeding their capacities. This check involves iterating through the weights and ensuring that the sum of weights does not exceed $NUM_LORRIES$ times $MAX_CAPACITY$. This operation has a complexity of $O(n)$.
3. Backtracking Process: The core of the algorithm is the backtracking process, where it recursively explores all possible combinations of assigning goods to lorries. At each step, it tries to place the current good in each lorry, branching into multiple recursive calls. The maximum depth of the recursion tree is bounded by the number of goods n , and at each level, there are $O(NUM_LORRIES)$ choices. Therefore, the time complexity of the backtracking process is $O(NUM_LORRIES^n)$.
4. Solution Output: After finding a feasible distribution, the algorithm outputs the assigned goods for each lorry along with the total cost. This operation involves iterating through the selected goods and has a complexity of $O(n)$.

Considering these components, the overall complexity of the backtracking algorithm is influenced primarily by the backtracking process, which has a complexity of $O(NUM_LORRIES^n)$. Additionally, the sorting operation contributes $O(n \log n)$, and the feasibility check and solution output operations contribute $O(n)$ each.

Thus, the total complexity of the algorithm can be described as:

$$T(n, NUM_LORRIES = 3) \in O(n + n \log n + 3^n) \approx O(3^n)$$

where n represents the number of goods, $MAX_CAPACITY$ is the maximum load capacity of the lorries, and $NUM_LORRIES$ is the maximum number of lorries.

- Description of Quicksort Time Complexity on Average Case

$$T(n) = O(n) + 2T\left(\frac{n}{2}\right)$$

$$n = 2^i$$

$$2^i - 2 \cdot 2^{(i-1)} = 2^i$$

$$r^i - 2 \cdot r^{(i-1)} = 2^i$$

$$r - 2 = 0$$

From left hand side we have got that $r = 2$ and $\alpha = 1$.

From the right hand side we have got that $r = 2$ and $\alpha = 1$.

In brief, $r = 2$ and $\alpha = 2$.

So that we can construct our expression with degree $\alpha - 1$.

$$T(2^i) = 2^i \cdot A + B \cdot 2^i \cdot i$$

On variable change,

$$T(n) = (A + B \cdot \log_2 n) \cdot n$$

Where $\Omega(n)$ and $O(n \log n)$

- Test of the algorithm with increasing values of n and analysis of the time taken by the algorithm (*CPU time*). Bear in mind that the specific time taken by the algorithm may vary depending on the machine used to run the code and the *GCC compiler* optimization settings. We must disable the optimization flag to get the real time taken by the algorithm, otherwise the compiler will optimize the code and, therefore, the graph of *CPU time* will not be a representative approximation.
 - GPU: *AMD Ryzen 5 3500U with Radeon Vega Mobile Gfx*
 - GCC compiler: *gcc (Ubuntu 11.4.0-1ubuntu1 22.04) 11.4.0*
 - GCC optimization flag: *#pragma GCC optimize("O0")* (GCC compiler optimization disabled)

```

1 #include <time.h>
2 #pragma GCC optimize("O0")
3
4 if (!feasibilityCheck(weights, n, LOAD_CAPACITY, NUM_LORRIES, BUDGET, COST_PER_KG)) // O(n)
5 {
6     printf("\nIt's not possible to distribute all goods in 3 lorries without exceeding the
7     capacity of 20 Tm each.\n");
8 }
9 else
10 {
11     qsort(weights, n, sizeof(int), compare); // O(n log n)
12     bool result = canPlaceGoods(weights, lorries, 0, n, LOAD_CAPACITY); // O(3^n)
13 }
14 end = clock();
15 cpu_time_used = ((double)(end - start)) / CLOCKS_PER_SEC * 1000.0;

```

Listing 1: Optimization settings and time measurement.

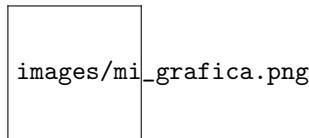


Figure 5: Graphic showing the cost of each strategy compared within each dimension of the problems.

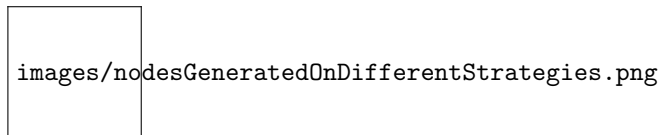


Figure 6: Density graphic showing the nodes generated of each Search Strategy.

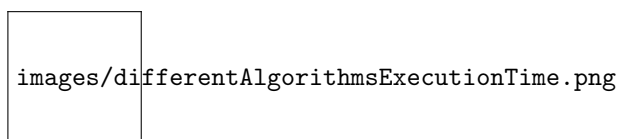


Figure 7: Negative execution time for A* strategy on huge dimension of problems.

- **Conclusion:** The empirical CPU time used by the optimized backtracking algorithm exhibits an unexpected trend when compared to the predicted exponential complexity of $O(3^n)$. The graph suggests a performance that is closer to linear, possibly due to effective optimizations. This discrepancy between theoretical and observed complexity underscores the impact of practical enhancements in algorithm implementation. Assuming the number of operations is proportional to the number of items, the empirical complexity could be approximated as $O(n)$, a significant improvement over the theoretical expectation.

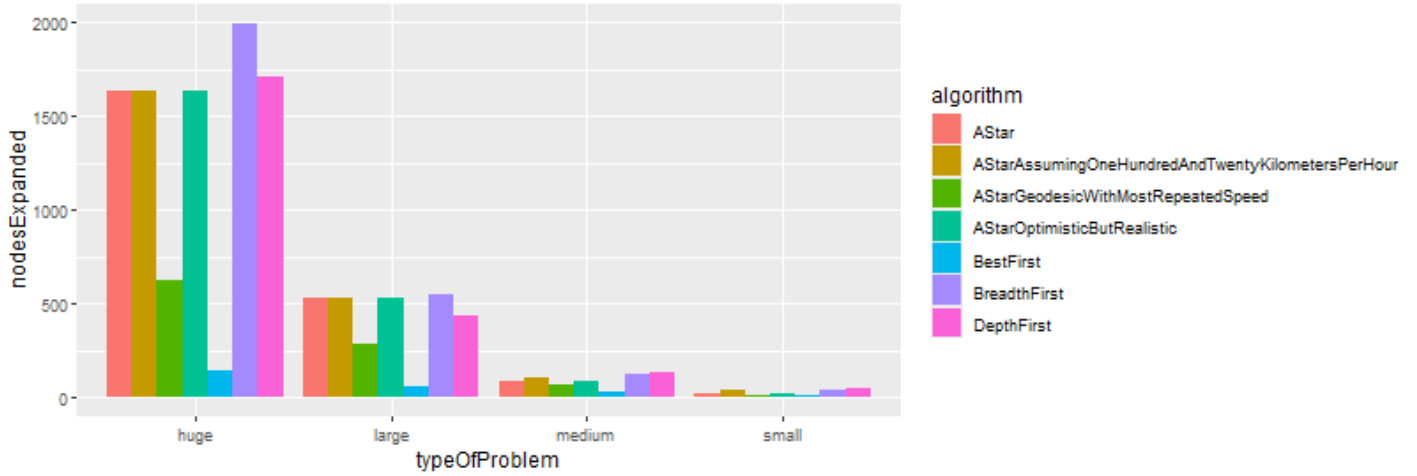


Figure 8: Average Nodes Generated on different search strategies across problems with different dimensionality.